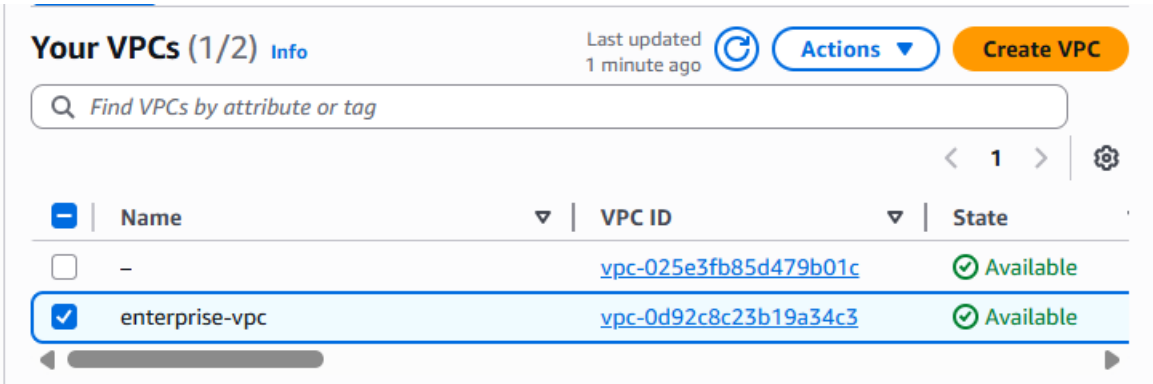
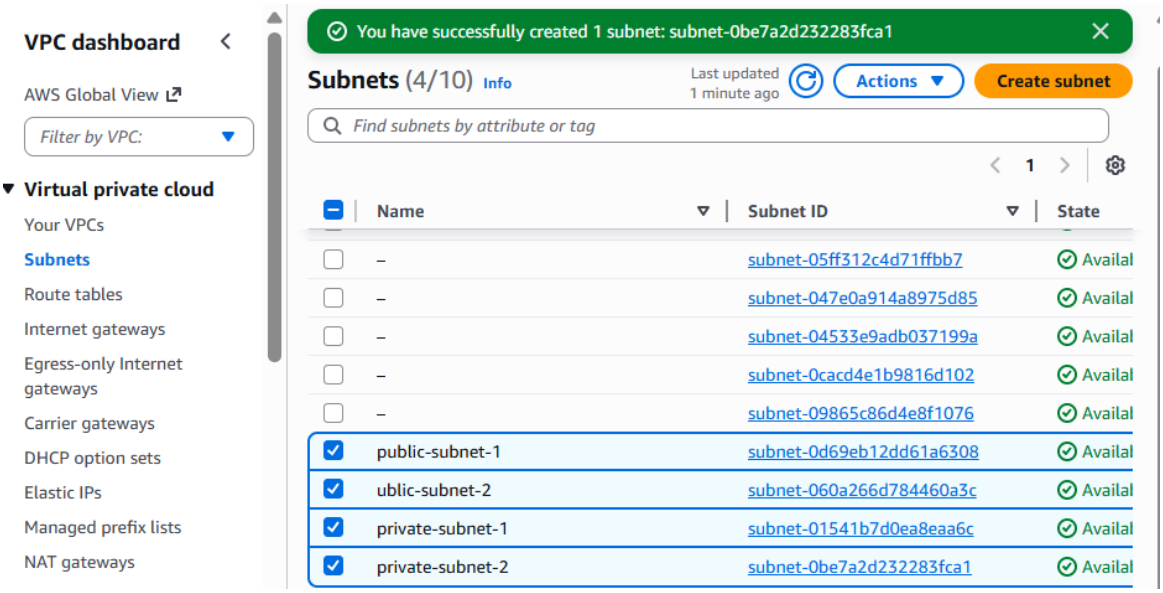
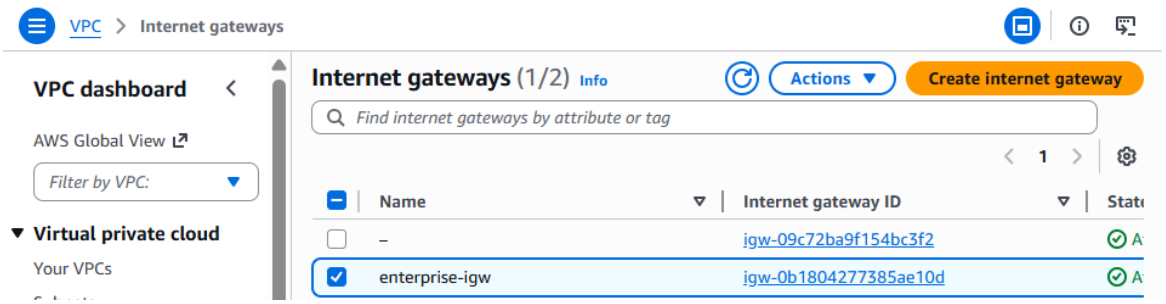
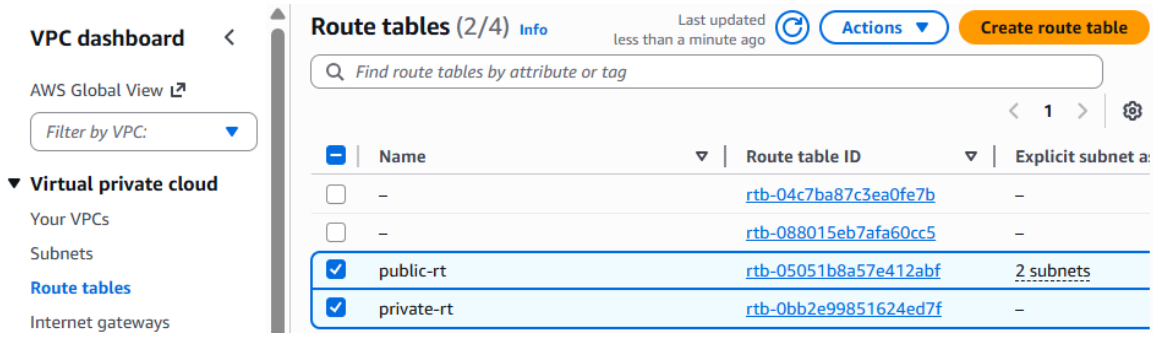
Networking Layer (VPC Design)

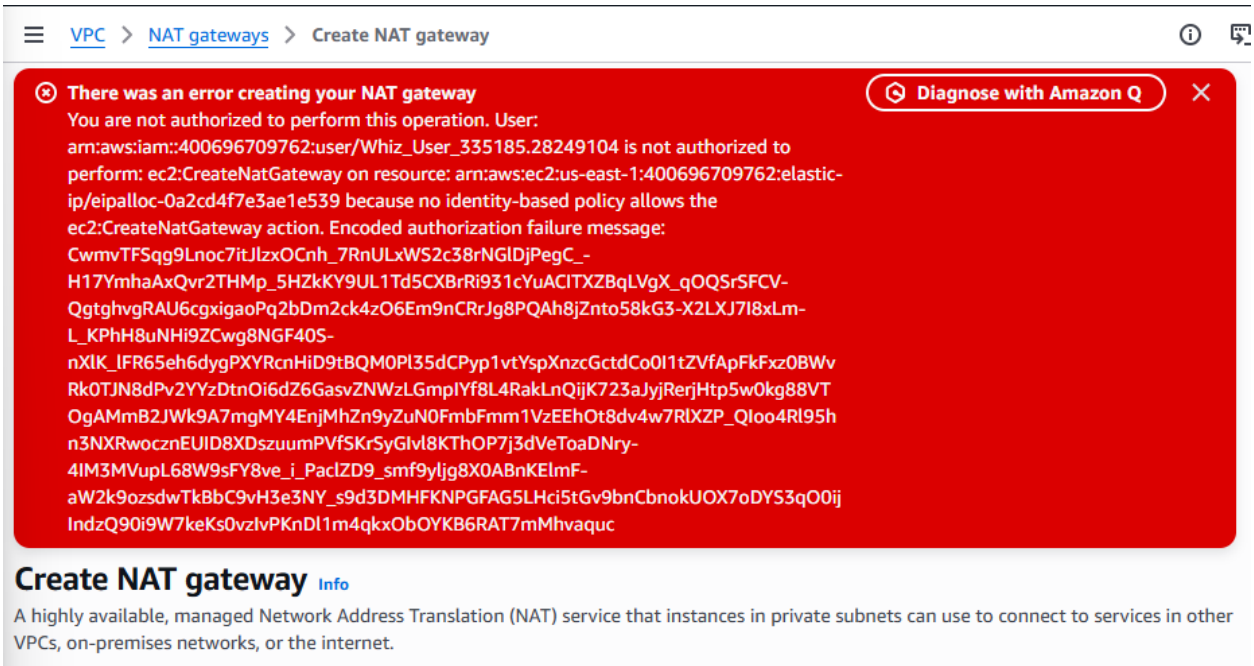




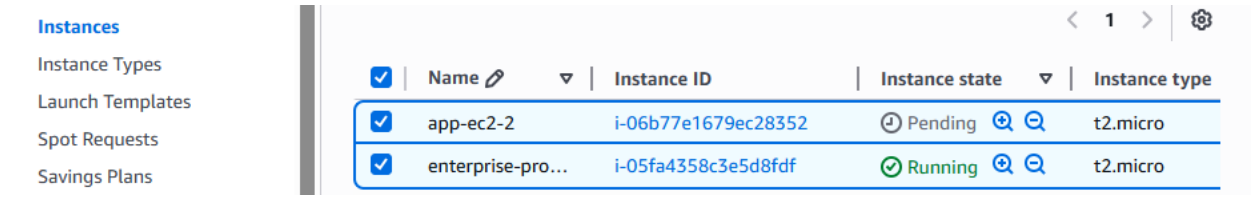




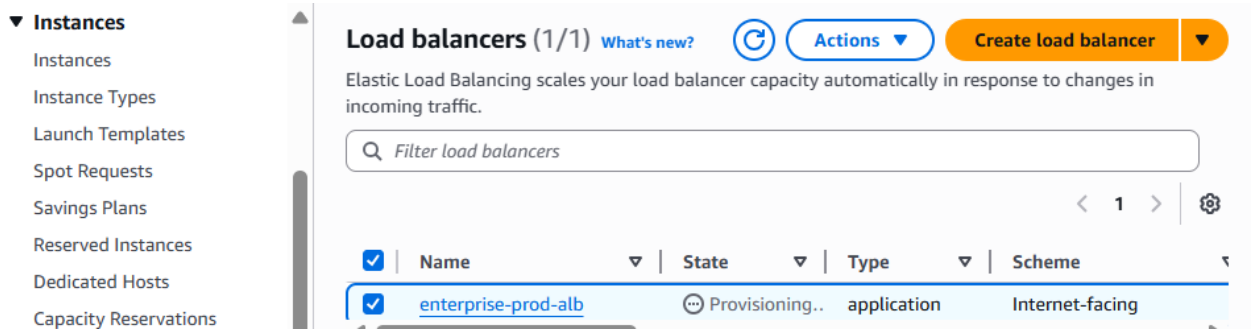
Due to sandbox environment limitations, NAT Gateway creation was restricted. In a real production environment, NAT Gateway would be used to allow private instances to access the internet securely.



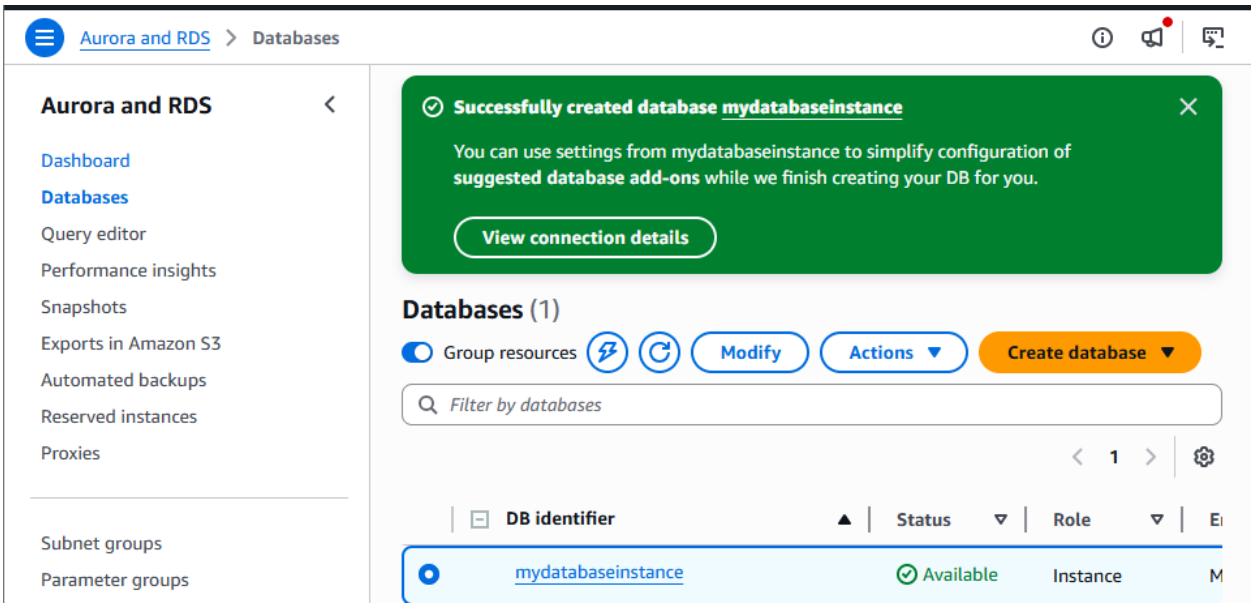
Compute Layer



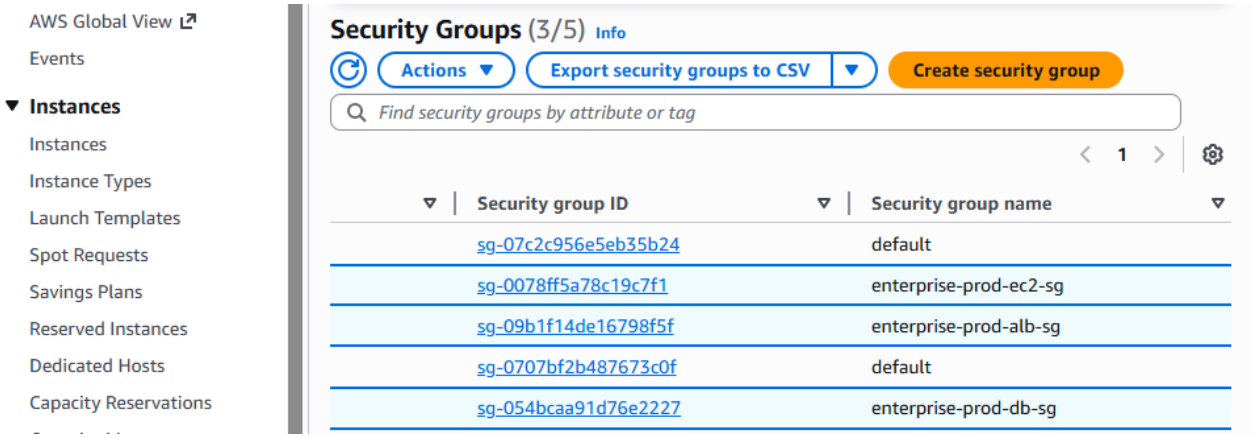
Load Balancing



Data Layer

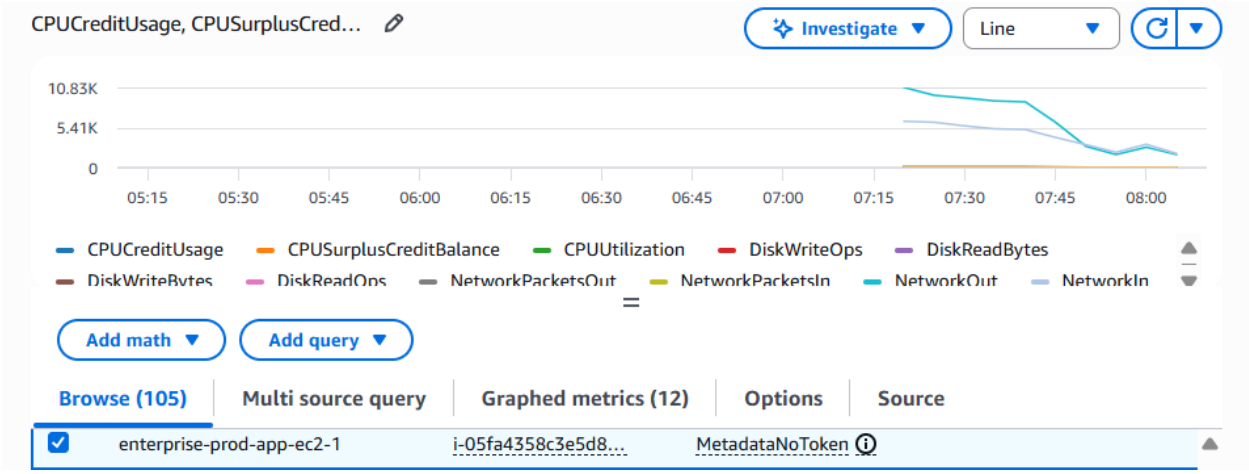


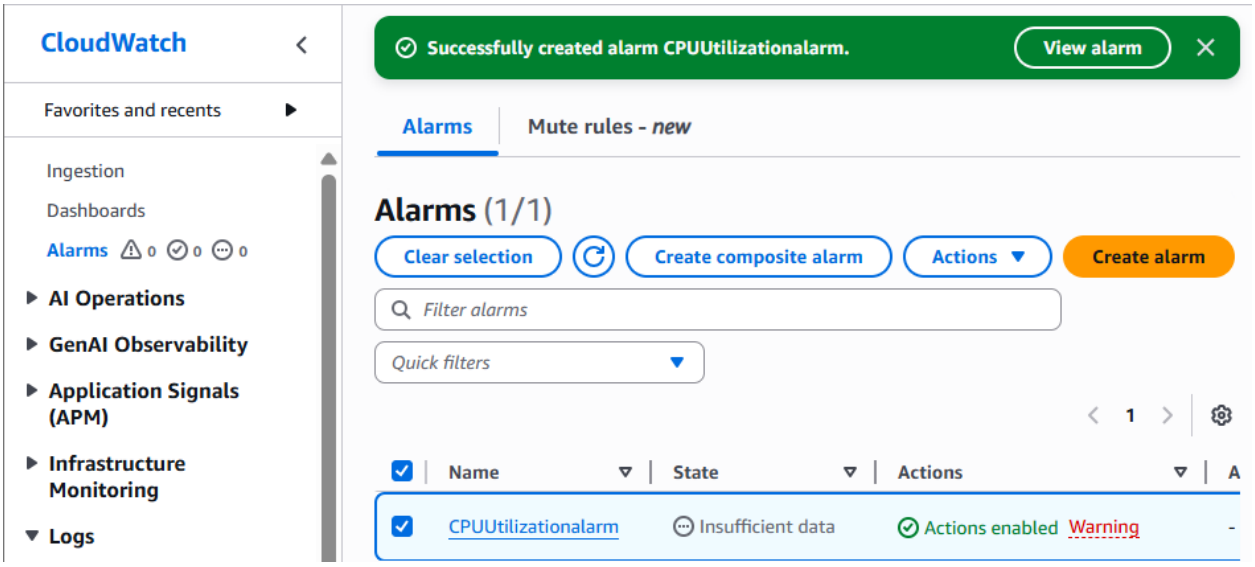
Security



Monitoring & Logging

Here we can monitor all servers





Automation (Mandatory)

VPC_template.json | VPC_II_template.json | Release Notes: 1.116.0

C: > Users > user > Downloads > VPC_II_template.json > ...

```
1  {
2      "AWSTemplateFormatVersion": "2010-09-09",
3      "Description": "Deploy a VPC",
4      "Resources": {
5          "VPC": {
6              "Type": "AWS::EC2::VPC",
7              "Properties": {
8                  "CidrBlock": "10.0.0.0/16",
9                  "EnableDnsHostnames": true,
10                 "Tags": [
11                     {
12                         "Key": "Name",
13                         "Value": "Lab VPC"
14                     }
15                 ]
```

VPC_template.json | VPC_II_template.json | Release Notes: 1.116.0

C: > Users > user > Downloads > VPC_template.json > { } Resources > { } AttachGateway

```
1  {
2      "AWSTemplateFormatVersion": "2010-09-09",
3      "Description": "Deploy a VPC",
4      "Resources": {
5          "VPC": {
6              "Type": "AWS::EC2::VPC",
7              "Properties": {
8                  "CidrBlock": "10.0.0.0/16",
9                  "EnableDnsHostnames": true,
10                 "Tags": [
11                     {
12                         "Key": "Name",
13                         "Value": "Lab VPC"
14                     }
```

CloudFormation > Stacks

Stacks (1)

Delete stack | Update stack | Stack actions | Create stack

Search by stack name

Filter status

Active

View nested

1

Stack name	Status	Created time	Descript
MyStack123	CREATE_COMPLETE	2026-04-20 13:30:21 UTC+0400	Deploy a

MyStack123

Delete stack | Update stack | Stack actions | Create stack

Stack info | Events | Resources | Outputs | Parameters | Template | Changeset

Outputs (3)

Search outputs

Key	Value	Description	Export name
AZ1	us-east-1a	Availability Zone 1	-
AZ2	us-east-1b	Availability Zone 2	-
VPC	vpc-00d22149e1968346f	VPC	-

VPC > Subnets

VPC dashboard

AWS Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections

Subnets (4/10) Info

Last updated less than a minute ago

Actions | Create subnet

Find subnets by attribute or tag

Name	Subnet ID	State
Private Subnet 1	subnet-0325ee9547fa57f8f	Available
Public Subnet 2	subnet-06c730641850eef95	Available
–	subnet-0fcef35f46f72cb1e	Available
Private Subnet 2	subnet-03c634b078f1d257e	Available
–	subnet-0dd41a895898d8881	Available
–	subnet-0e2a0e7e779f55616	Available
–	subnet-0e3cecfb49bbcc227	Available
–	subnet-0ac3cd497dfff380e	Available
Public Subnet 1	subnet-0113a0e7be97d46d0	Available
–	subnet-03e3e88d1df5e5459	Available